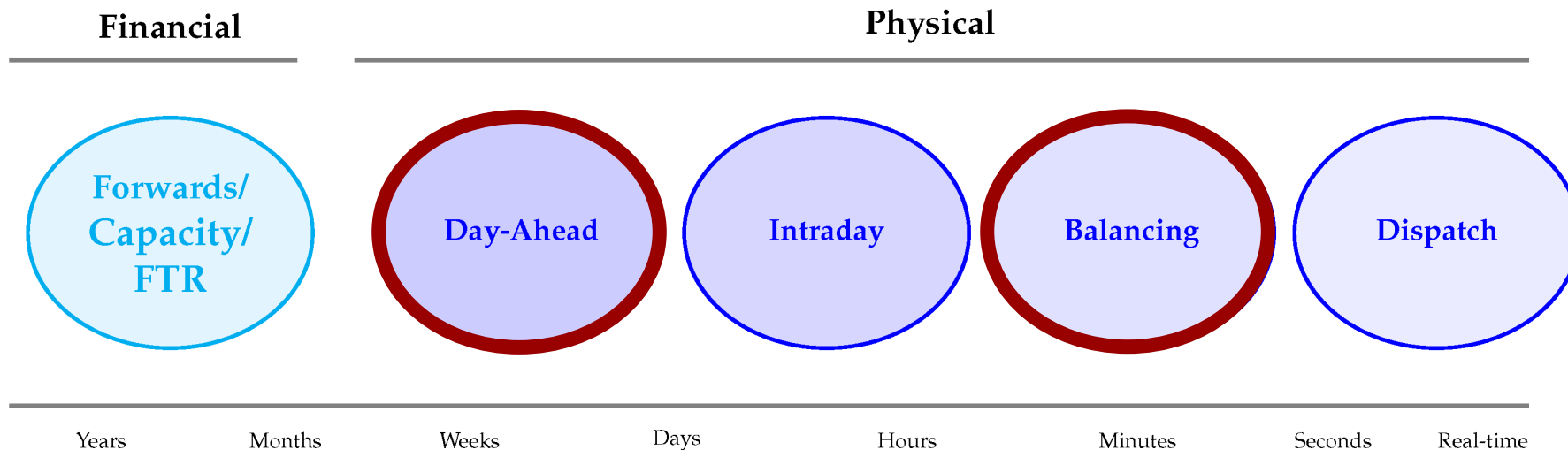


Cooperative Redistribution of Imbalance Cost
Savings

Sharing the Gains of Aggregation

The Problem

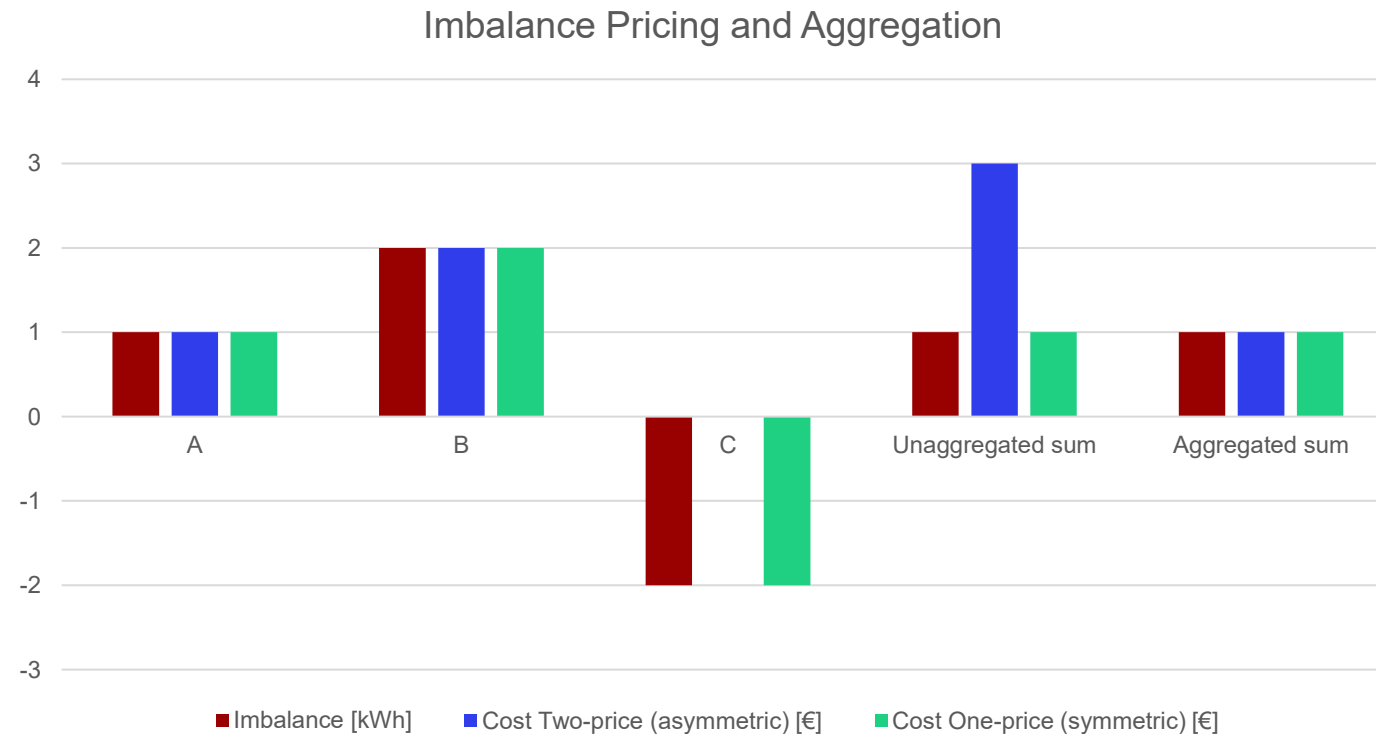
- Power consumption must always match power purchased
- Trading is done in day-ahead and intraday markets by Balancing Responsible Parties based on forecasts, mismatches are resolved in the balancing market
 - This thesis only models the day-ahead and balancing market
 - Participating in the balancing market carries financial and regulatory risk



Source: O'Connor et al. (2025), *Energies*, CC BY.

Asymmetric VS Symmetric Imbalance Pricing

- Symmetric pricing reward and punishes clients equally
- Asymmetric pricing never rewards or rewards less than it punishes
- Asymmetric pricing increases importance of high forecast accuracy



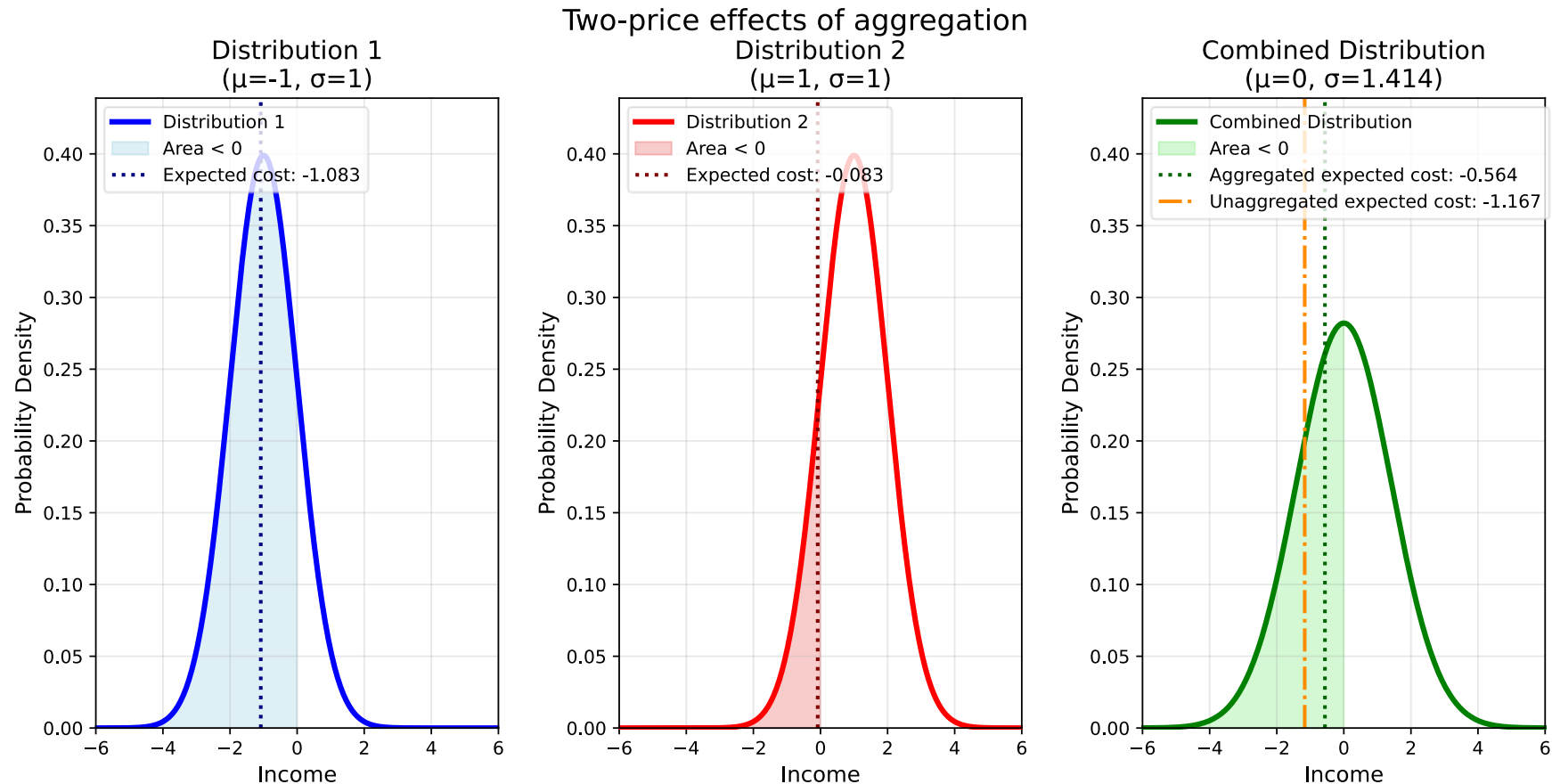
The Case

- Reel cooperates with a solar park in Mesballe
- 25 signatories (22 with usable data) have a Power Purchase Agreement
- Each has Reel as Balancing Responsible Party
 - Reel must forecast consumption and production, and trade the difference in electricity markets



The Cost problem

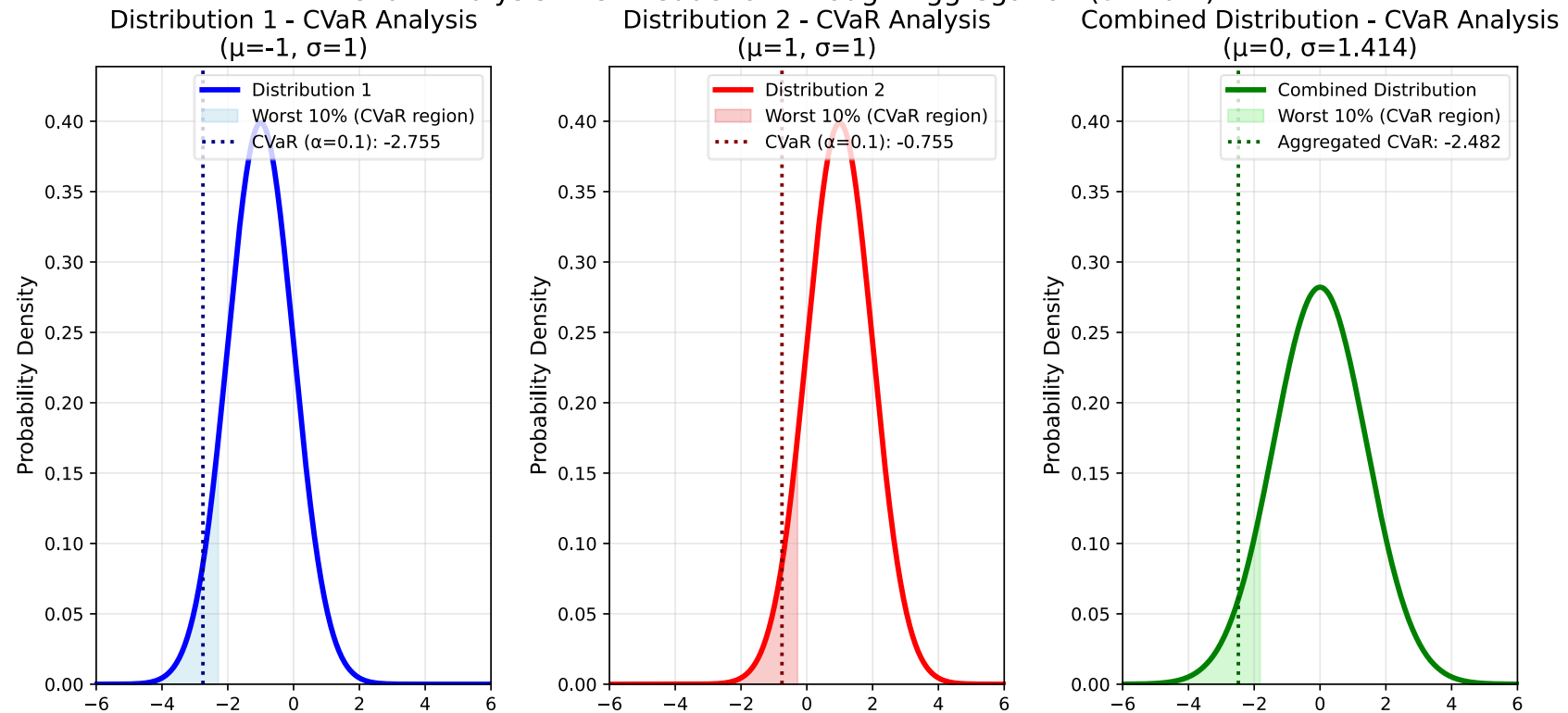
- Non-shaded areas have an income of 0 in a two-price scheme
- Aggregation reduces expected cost under asymmetric imbalance pricing



The Risk Problem

- Risk metric is Conditional Value at Risk (CVaR)
 - CVaR is the average income in the worst 10% (or other percentage) scenarios
- CVaR is reduced by aggregation as worst case scenarios do not always coincide

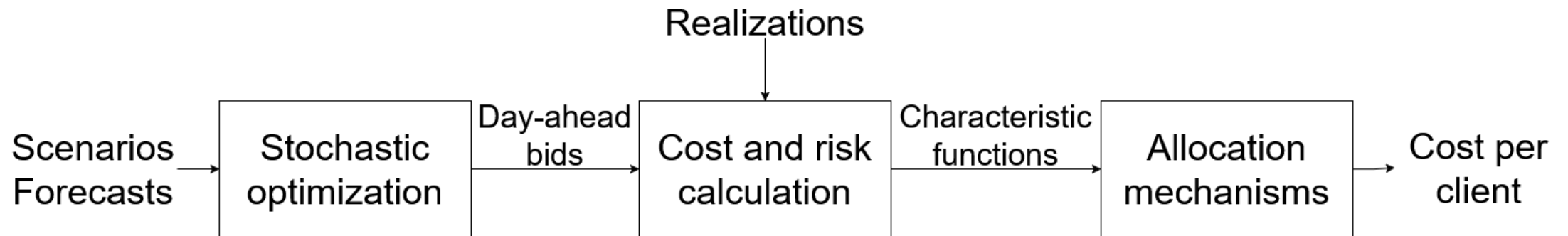
CVaR Analysis: Risk Reduction through Aggregation ($\alpha = 0.1$)



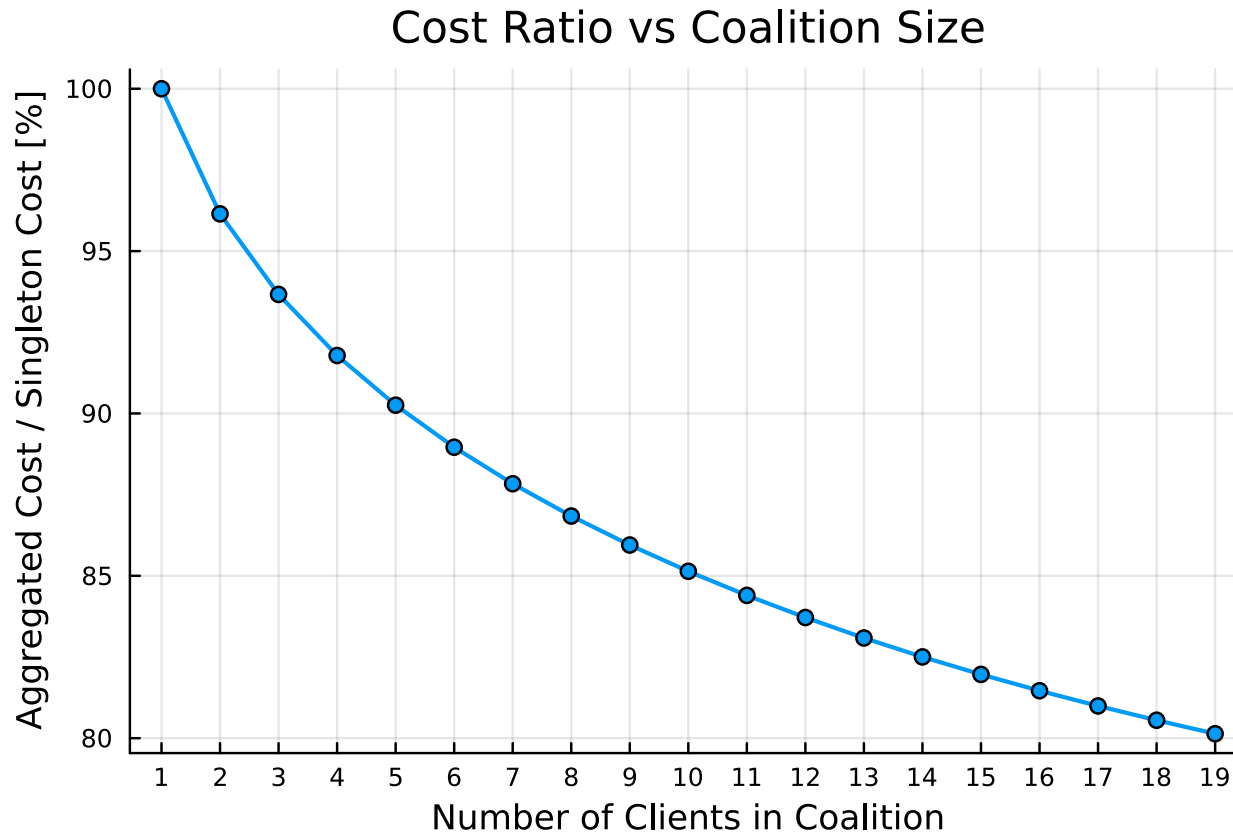
The Model

- Focus of the thesis has been on the allocation mechanisms
- Stochastic model input:
 - 5 sets of 1 week persistence forecast scenarios for demand
 - 30 imbalance price scenarios
 - Deterministic solar forecast
- Bids for clients are independent – cheap calculation of coalitions

**Characteristic
function:
The cost or risk of
a group of clients**



The Gains of Aggregation



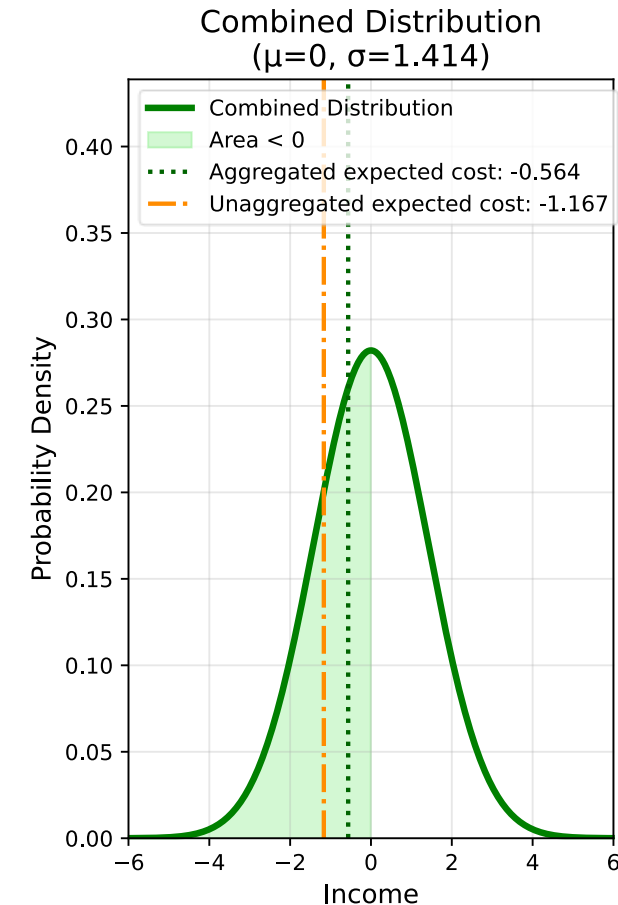
- In an electricity market with asymmetric imbalance pricing, aggregation of clients reduces costs and risk
- With symmetric pricing, aggregation only reduces risk

Total Aggregated Cost	Total Singleton Cost	Cost reduction
157617€	196756€	19.9%

Table 7.2: Reduction in costs from aggregation, 22 clients

The Case for Game Theory

- Clients cannot just pay their individual costs
 - This will cause overpayment
- A good “allocation” of costs will reflect client contribution
 - One possibility is to just scale individual costs, but there are other methods



Studied Mechanisms

The classics

- The Shapley Value
- Vickrey-Clarke-Groves (VCG)
- The Nucleolus
- The Gately Point (Both calculated for full period and for every 15 minutes)

Novel mechanisms

- Budget Balanced VCG
- Uniform Price
- Reduced Cost

Benchmark

- Flat Rate

Choosing the best allocations

- The maximum excess indicates how much a coalition of client could gain by leaving the cooperation
- A negative maximum excess is stable
- Most allocations are stable or close to stable

Clients	Shapley	VCG	VCG BB	Gately Point	Gately 15 min	Uniform Price	Reduced Cost	Flat Rate	Nucleolus
19	4.7€	-132.2€	-0.1€	-36.2€	-16.0€	-16.2€	2.6€	15105€	–
12	-63.0€	-789.7€	-0.6€	-220.9€	-188.0€	-81.9€	98.0€	10868€	-394.8€

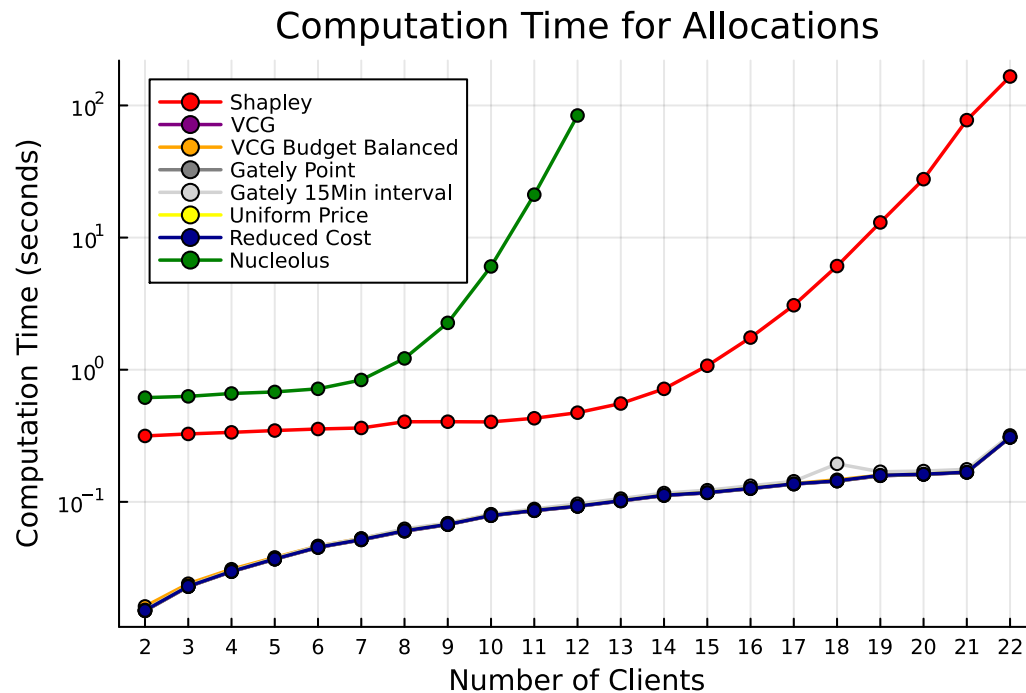
Table 7.3: Maximum excess of allocations for cost

Clients	Shapley	VCG	VCG BB	Gately Point	Uniform Price	Flat Rate	Nucleolus
19	0.02€	-0.21€	0.001€	-0.058€	-0.0005€	14.9€	–
12	-0.08€	-1.04€	-0.0005€	-0.29€	-0.0014€	10.72€	-0.52€

Table 7.7: Maximum excess of allocations for CVaR

Choosing the best allocations

Computation time: Nucleolus and Shapley scale quickly all others very similar



Budget balance: VCG will always make clients underpay.

Total Cost	Total VCG Cost	Unassigned Cost
157617€	142919€	9.3%

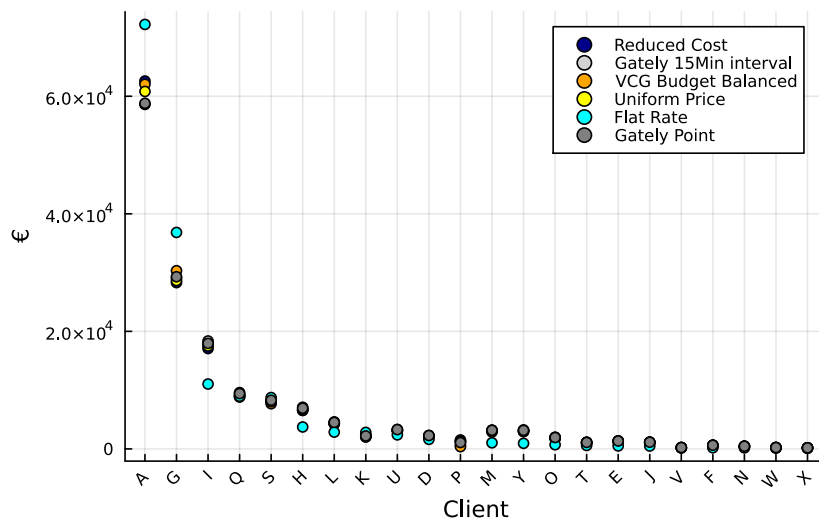
Table 7.5: Budget balance of VCG for cost problem, 22 clients

Aggregated CVaR	Sum of CVaR	Difference
180.3€	159.9€	11.3%

Table 7.6: Difference between individual and aggregated CVaR, 22 clients

Cost Contribution

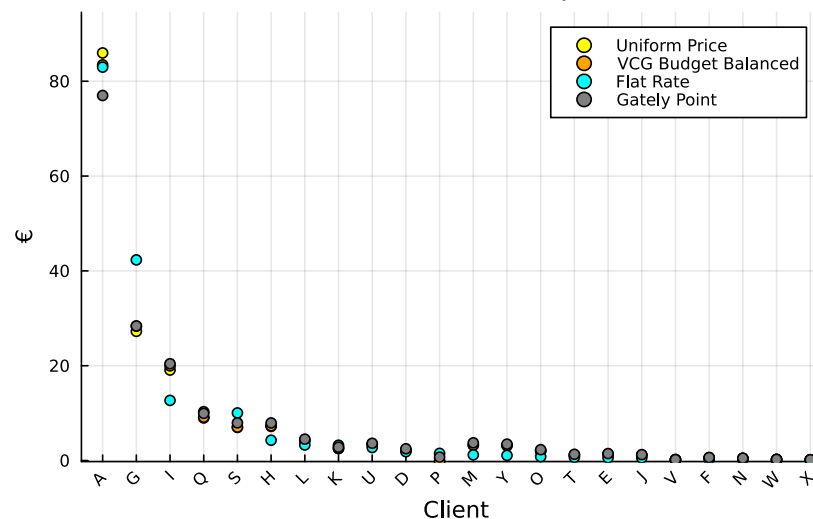
Total imbalance cost per client



- Closely follows MWh demand
- Client A and G overpay with a flat rate
- Cost and CVaR closely related

CVaR Contribution

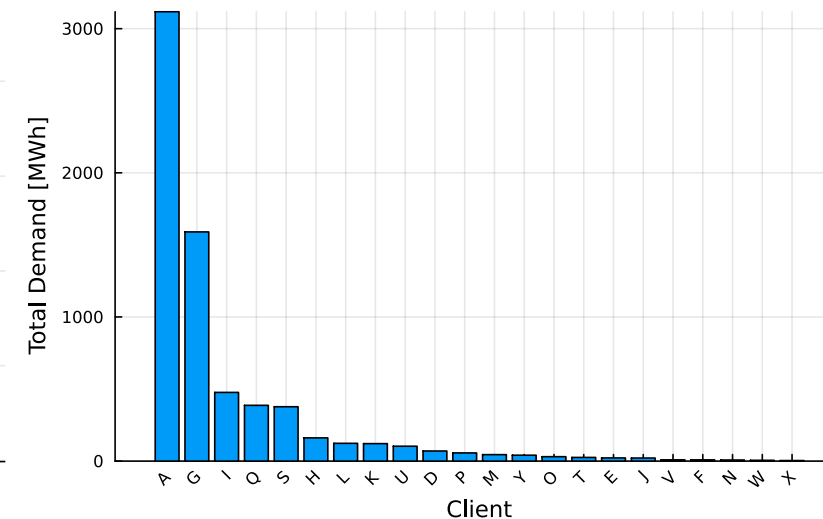
CVaR contribution total per client



- Closely follows MWh demand
- Client G overpays with a flat rate
- Reduced Cost and Gately 15Min not implemented for CVaR

Total Demand

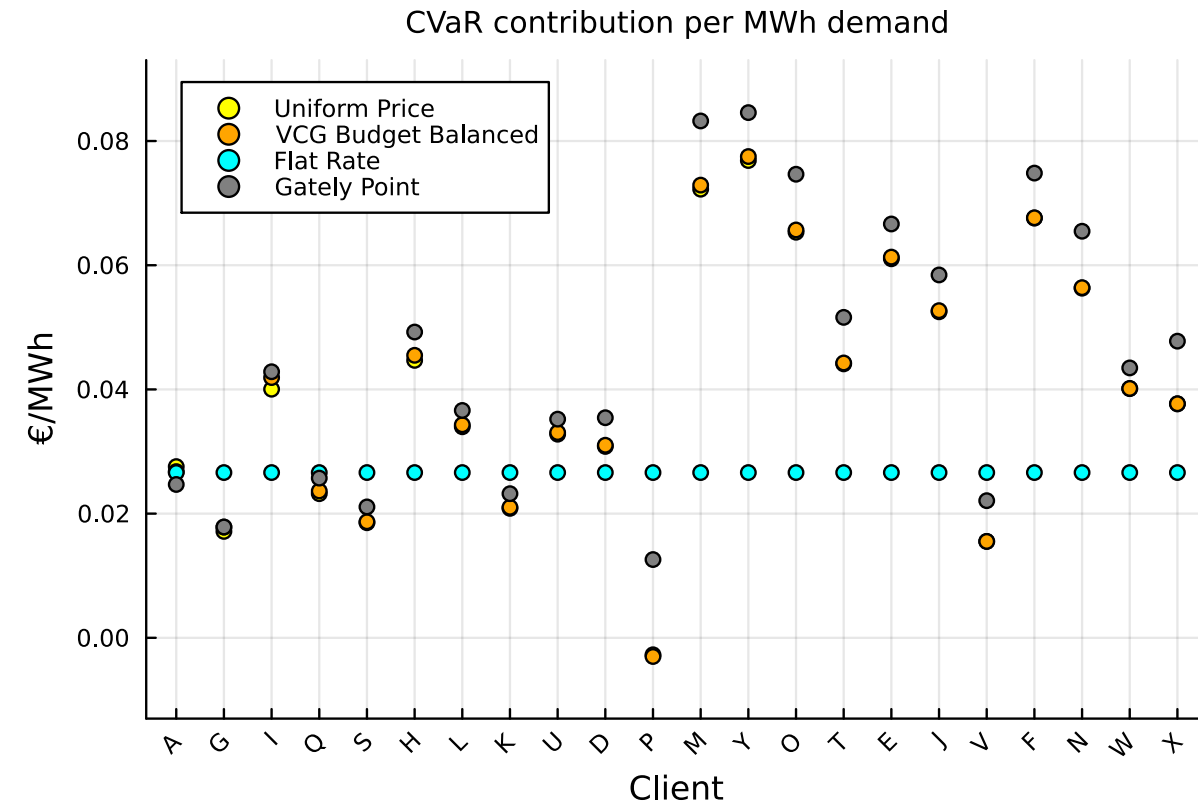
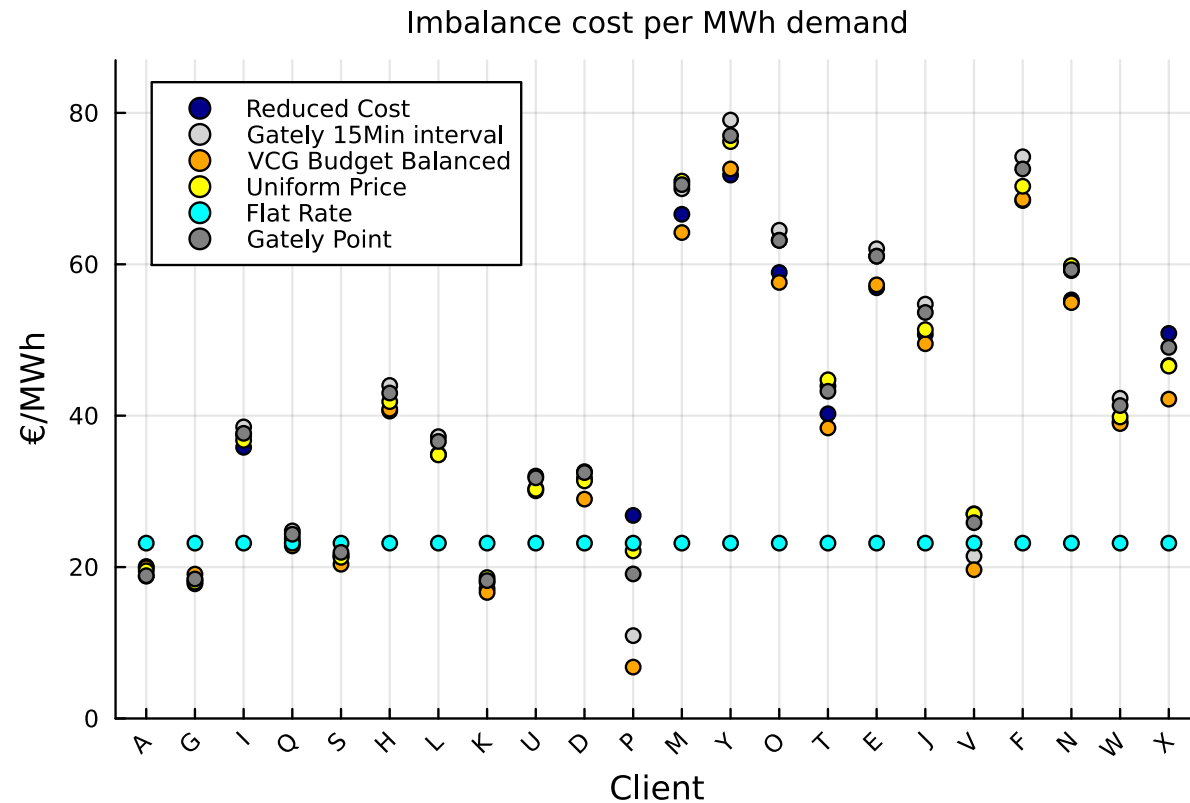
Total MWh Demand per Client



- Large difference in client size
- Client A has larger demand than the rest combined

Allocation Per MWh

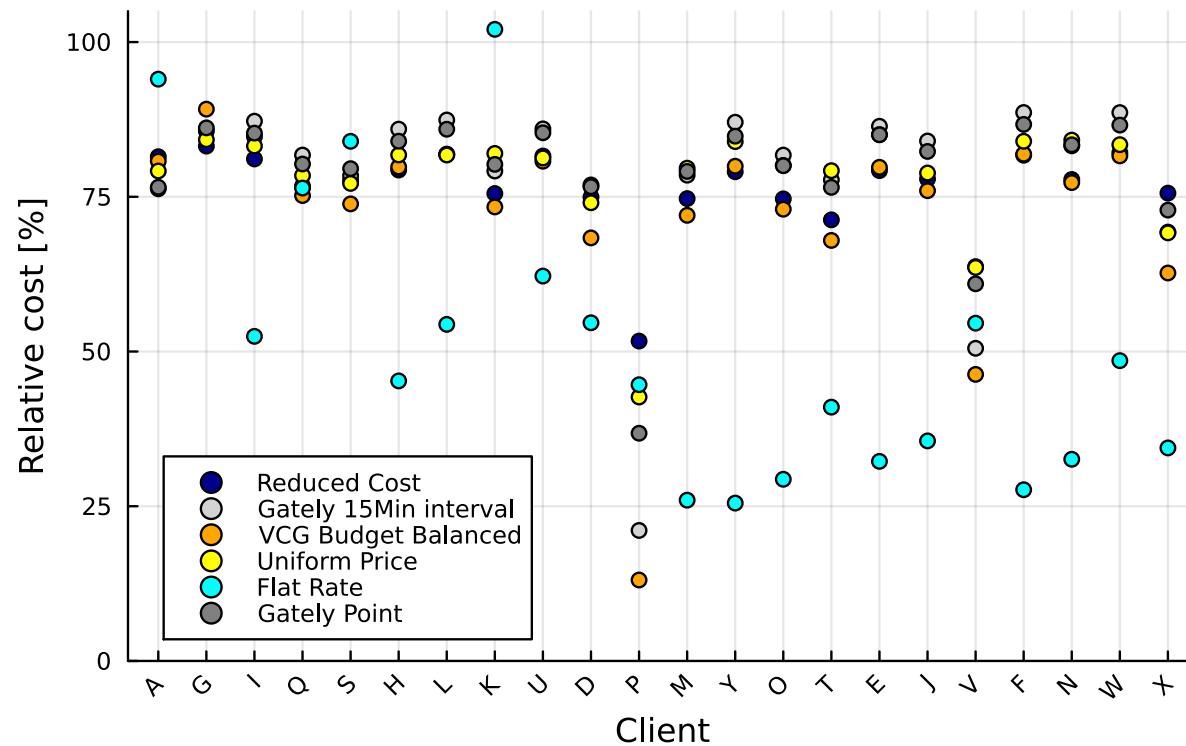
This indicates the fee that each client should pay to compensate for the cost they add



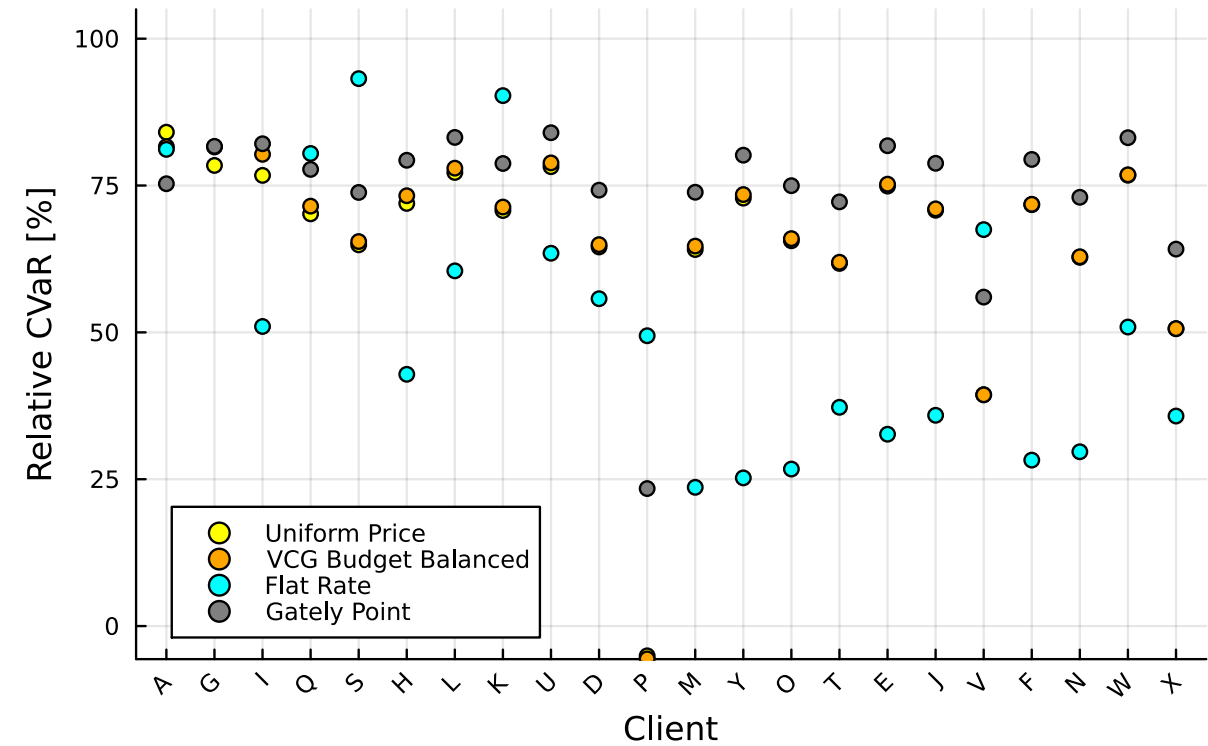
Allocation Compared To Singleton Cost

This indicates the cost and risk attributed to each client compared to their cost alone

Imbalance cost contribution vs individual cost



CVaR contribution vs individual CVaR



Concluding

Large variation in client costs indicate the need for individual pricing

- Many clients have similar relative costs – an individual assessment might be enough

The Gately Point, Reduced Cost and Uniform Price are the best mechanisms

- Gately point is a classical mechanism that simulates market power
- Reduced cost and uniform price are simpler for clients to understand

The difficulty in this isn't the game theory, it is producing accurate characteristic functions

- Requires somehow approximating intraday markets or disregarding them
- Requires access to client data ex-ante or a product where costs are calculated ex-post

DTU



Thank you for your attention

Studied mechanisms

Mechanism	Formula	Computational complexity	Budget balanced	Group rational	Additive	Incentive compatible
Shapley Value	$x_i = \sum_{S \in 2^N \setminus \{A_i\}} \frac{(S)!(N - S -1)!}{N!} (v(S \cup \{A_i\}) - v(S))$	XXXX	Yes	No	Yes	Yes*
VCG	$v(N) - \sum_{j \in N \setminus \{i\}} v_j(N \setminus \{i\})$	X	No	Yes	Yes	Yes
VCG Budget Balanced	Minimize squared distance to VCG allocation	XX	Yes	No	No	No
Uniform Price	$x_i = \sum_{t \in T} \begin{cases} -k_t^{\text{uni}} \cdot \Delta_{t,i}^{\text{Real}} , & \text{if } v(\{i\}) = 0. ** \\ v_t(\{i\}), & \text{otherwise.} \end{cases}$	X	Yes	Yes***	Yes	No
Reduced Cost	$x_i^N = \sum_{t \in T} \frac{v_t(\{i\})}{\sum_{i \in N} v_t(\{i\})} \cdot v_t(N)$	X	Yes	No	Yes	Yes*
Nucleolus	$\theta_i(x) = \epsilon(S_i, x)$, where $i = 1, \dots, 2^{ N } - 1$ $x = \{y \in I \mid \theta(z) \succeq_t \theta(y) \forall z \in I\}$	XXXXXX	Yes	Yes	No	Yes*
Gately Point	$d_i(x) = \frac{\sum_{j \neq i} x_j - v(N \setminus \{i\})}{x_i - v(\{i\})}$, $x = \arg \min_x \left(\max_i d_i(x) \right)$	XX	Yes	No**	No	Yes*

Table 7.1: Studied mechanisms and their formulas and properties

* Assuming true costs/utilities for everyone is known

** Slight difference between cost and risk allocation, see section 5.2.2

*** Group rational only for risk allocation

- In addition:
 - Gately point both calculated for full period and every 15 minutes to show effect of lacking additivity
 - “Flat rate” benchmark allocation where everyone pays the same

Cooperative Game Theory

Efficient solution

Player B

Cooperate Defect

Player A

Cooperate Defect

Cooperate	A: €1000 B: €1000	A: €10000 B: €0
Defect	A: €0 B: €10000	A: €5000 B: €5000

Fines given

Nash-equilibrium

- The Prisoners Dilemma
 - Players can choose to cooperate or defect
 - Goal is to minimize fine
- The cooperative solution requires agents to join a “coalition” together
- There are often infinite fair ways to divide costs in a coalition, core-selection mechanisms find specific fair allocations

Important concepts

- i , a client
- S , a coalition of clients: $i \in S$
- N , the grand coalition, the largest possible coalition $S \subseteq N$
- $v(S)$, the characteristic function of a game. The total cost of the coalition
- x_i^S , the vector of payments for clients $i \in S$

- Clients: A and B
- Possible coalitions: $\{A\}$, $\{B\}$, and $\{A,B\}$
- Grand coalition: $\{A,B\}$
- $v(\{A\})=5000$, $v(\{B\})=5000$, $v(N)=2000$

		Player A	
		Cooperate	Defect
Player B	Cooperate	A: €1000 B: €1000	A: €10000 B: €0
	Defect	A: €0 B: €10000	A: €5000 B: €5000
		Fines given	

Stability

- An **imputation** is an allocation that is individually rational and collectively rational.
- An imputation is in the **core** if it is also group rational.
- Individual rationality: $x_i^N \leq v(\{i\})$
- Collective rationality: $\sum_{i \in N} x_i^N = v(N)$
- Group rationality: $\sum_{i \in S} x_i^N \leq v(S)$
- $\epsilon(S, x^N) = \sum_{i \in S} x_i^N - v(S)$, the excess of clients. How much a coalition can benefit from splitting from the grand coalition
- The core:
- $x_A^N \leq 5000; x_B^N \leq 5000; x_A^N + x_B^N = 2000$

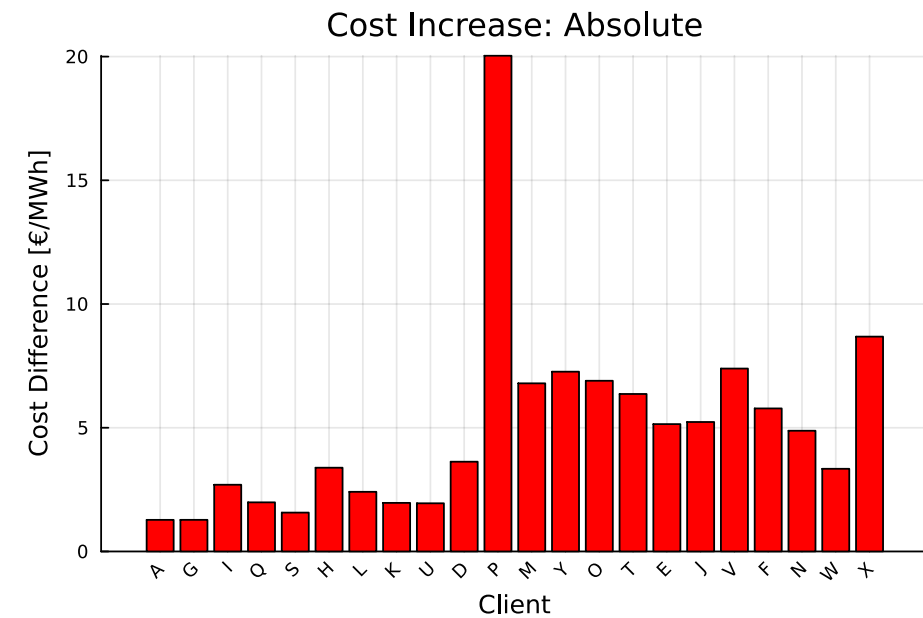
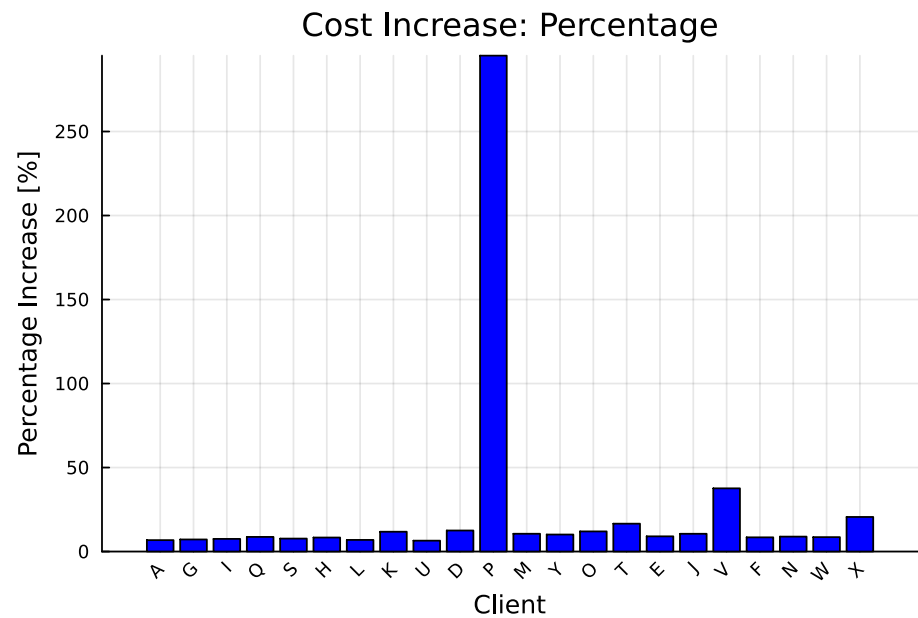
- Clients: A and B
- Possible coalitions: $\{A\}$, $\{B\}$, and $\{A,B\}$
- Grand coalition: $\{A,B\}$
- $v(\{A\})=5000$, $v(\{B\})=5000$, $v(N)=2000$

		Player A	
		Cooperate	Defect
Player B	Cooperate	A: €1000 B: €1000	A: €10000 B: €0
	Defect	A: €0 B: €10000	A: €5000 B: €5000
		Fines given	

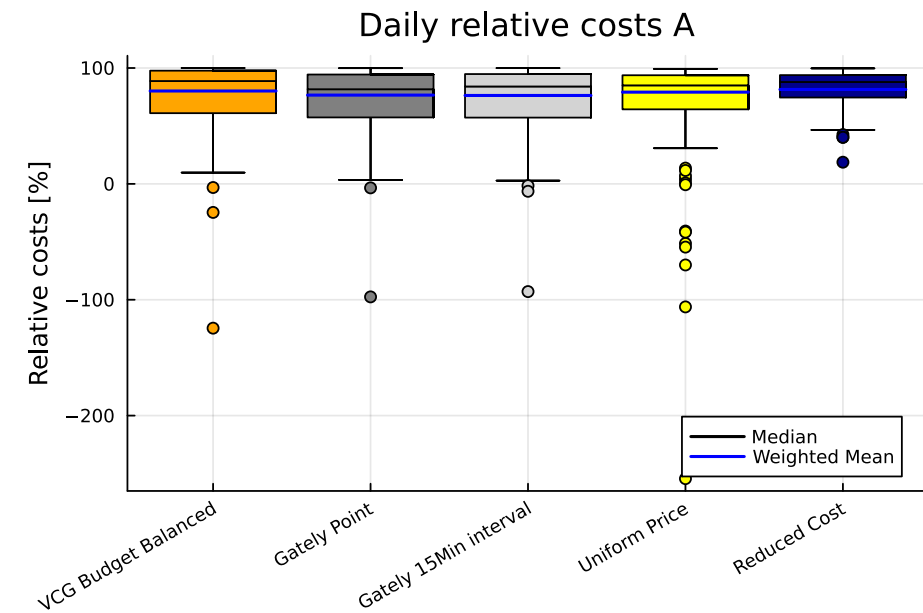
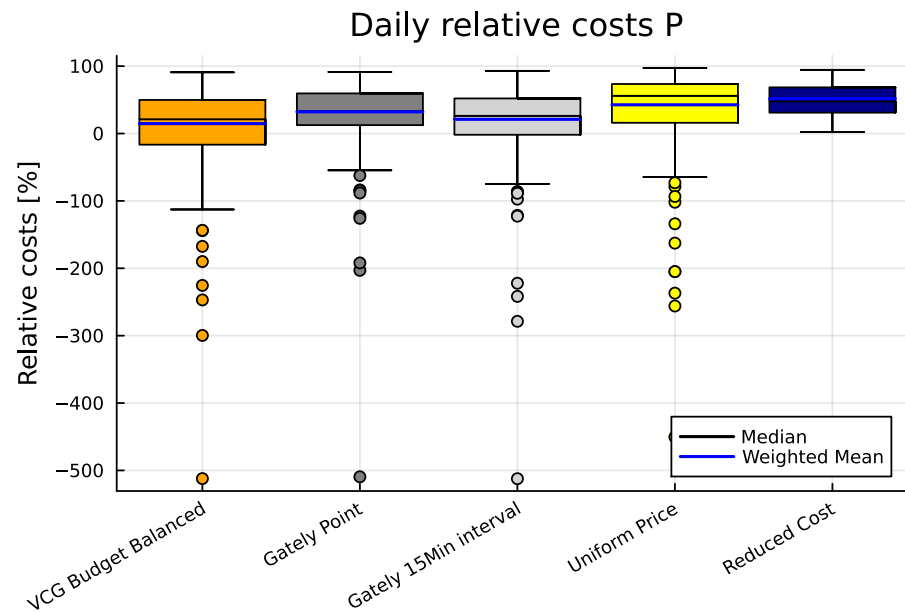
Desirable Properties of a mechanism

- **Incentive compatibility:** Every agent maximizes their own utility by acting truthfully
- **Budget balance:** The community manager does not have financial excess nor deficit.
- **Group rationality:** No group of agents can benefit from splitting from the community
- **Additivity:** The sum of allocations for a set of sub-games will be the allocation for the game created by adding together the sub-games

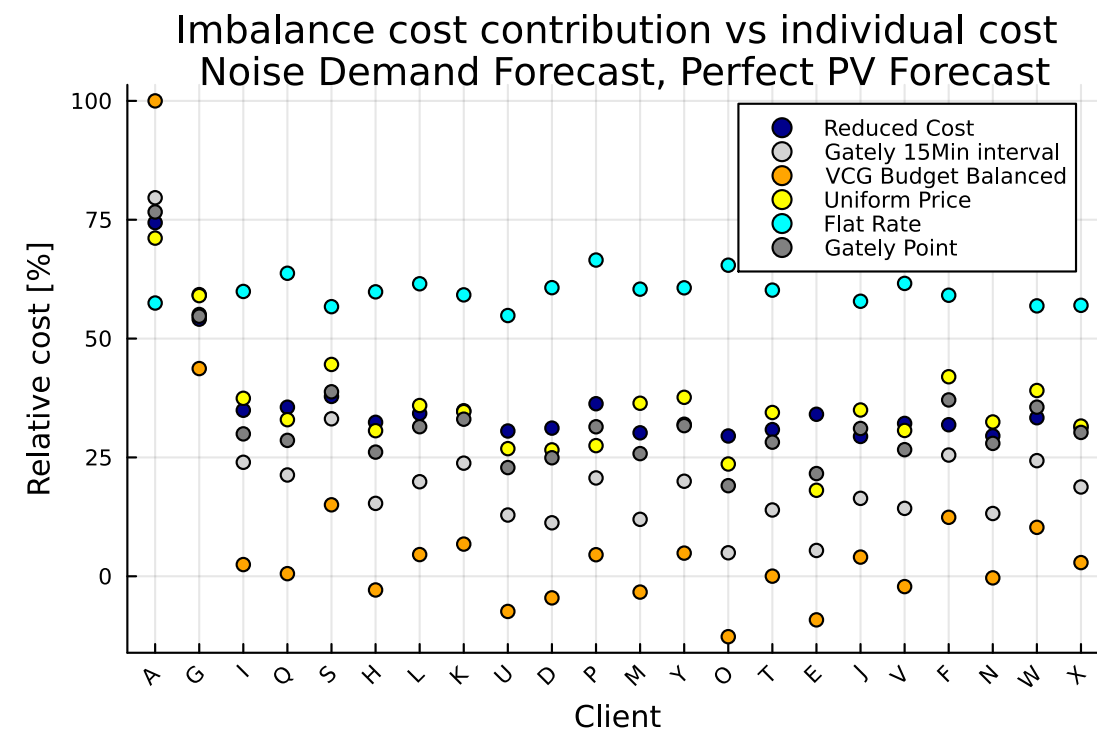
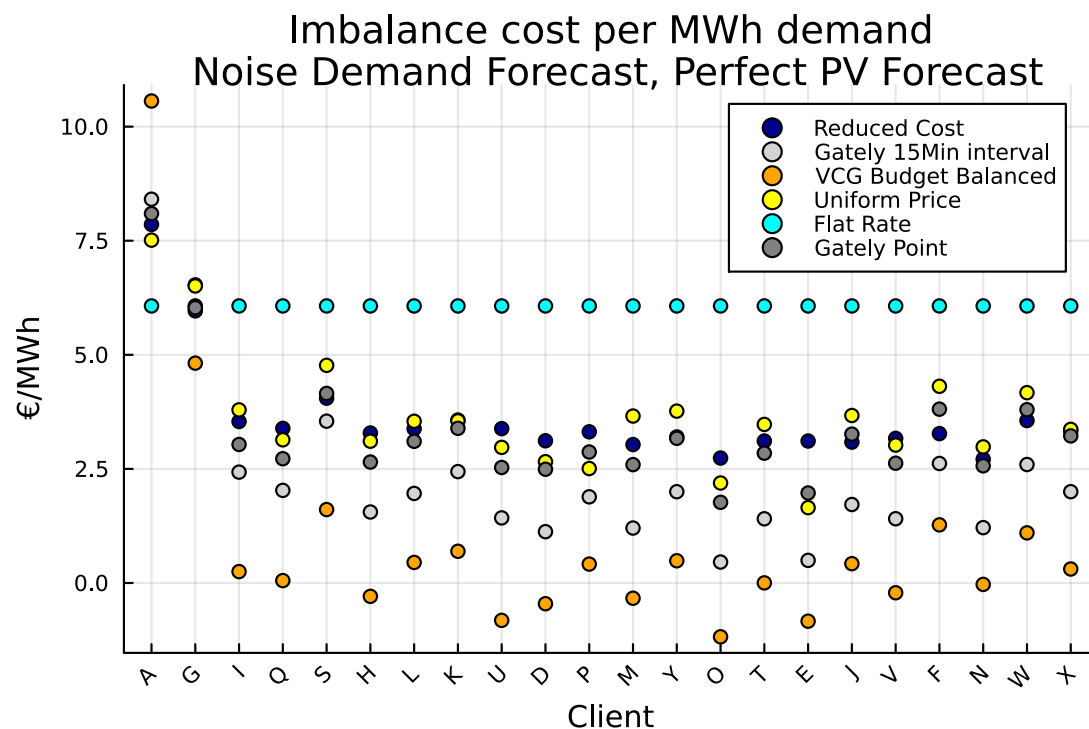
Cost increase between mechanisms



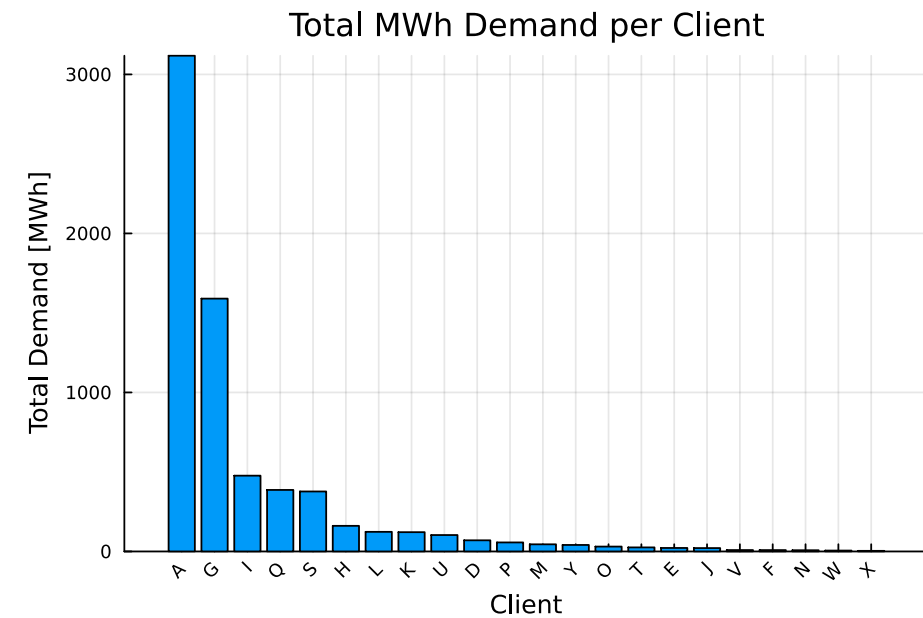
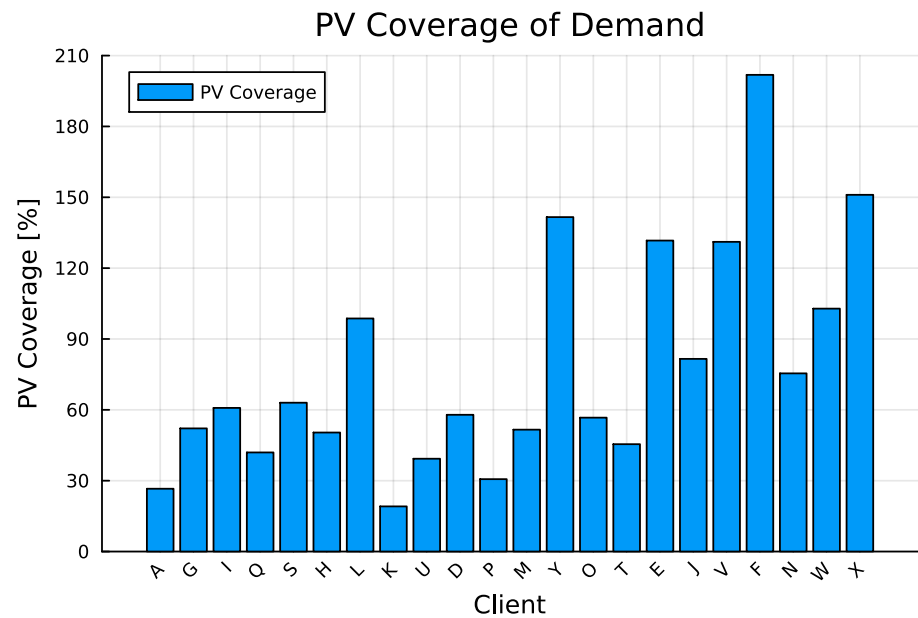
Daily variability



Better noise based “forecast”

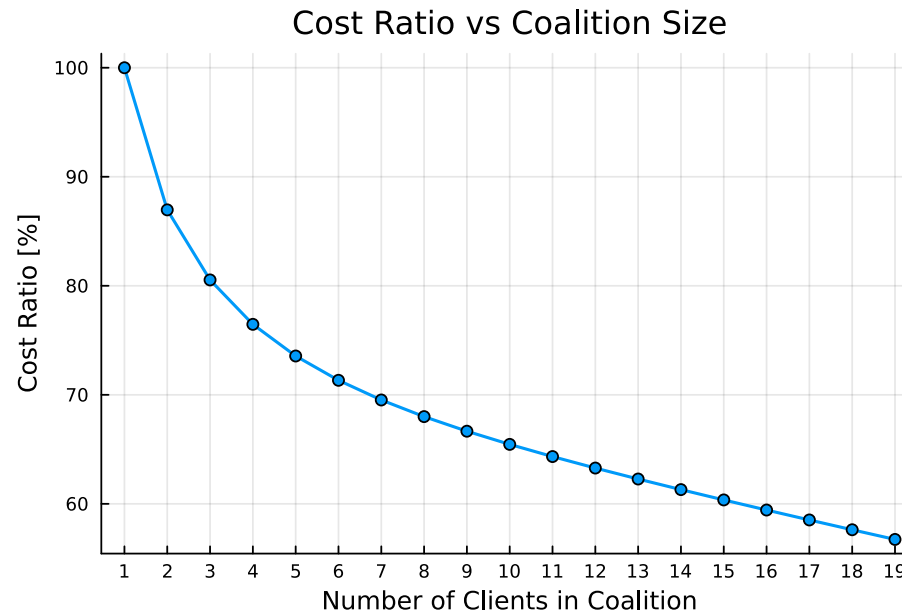


PV coverage of clients

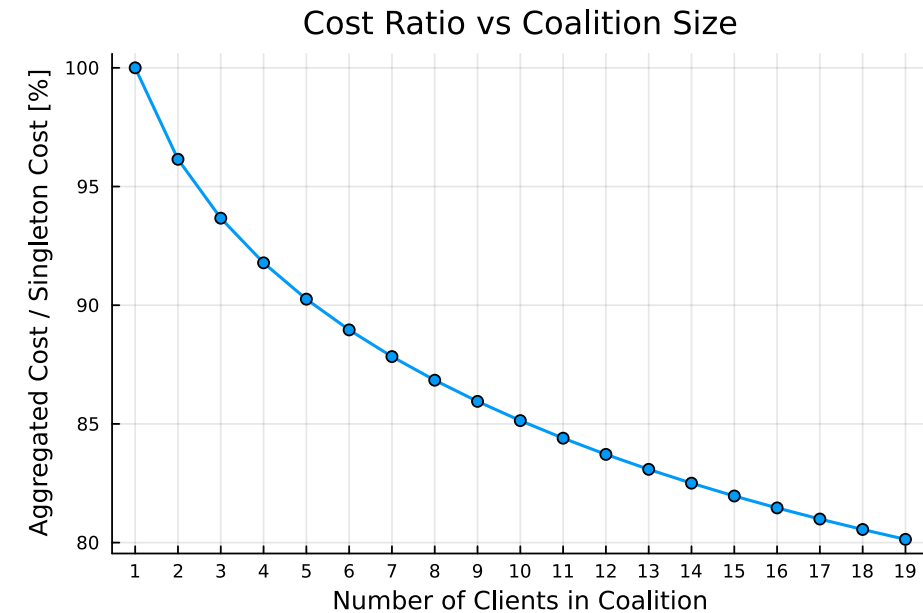


Comparing gains based on different forecasts

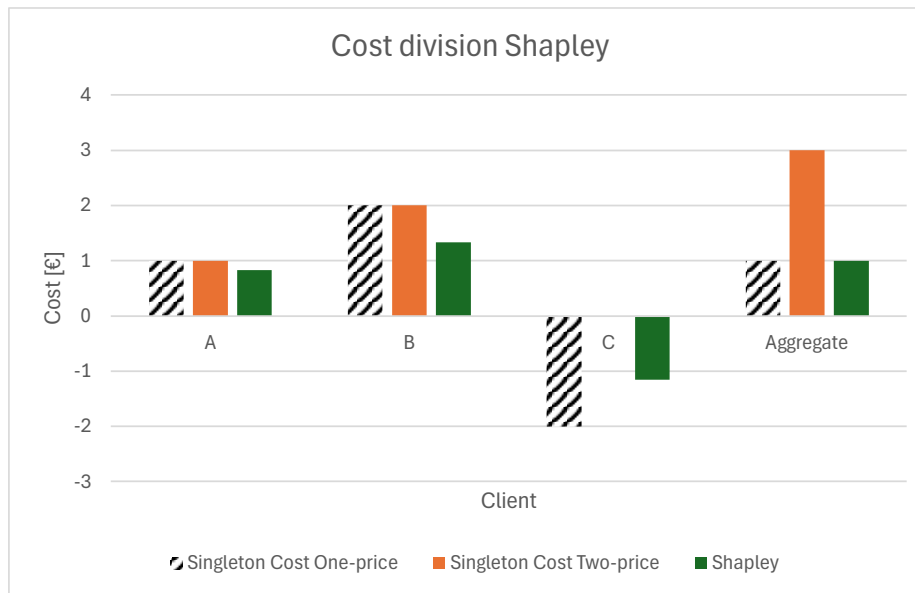
Using accurate noise based “forecast”



Scenarios based forecasting



Shapley allocation



$$\Phi_i(v) = x_i^N = \sum_{S \in 2^N \setminus \{i\}} \frac{(|S|)! (|N| - |S| - 1)!}{N!} (v(S \cup \{i\}) - v(S))$$

Uniform Price

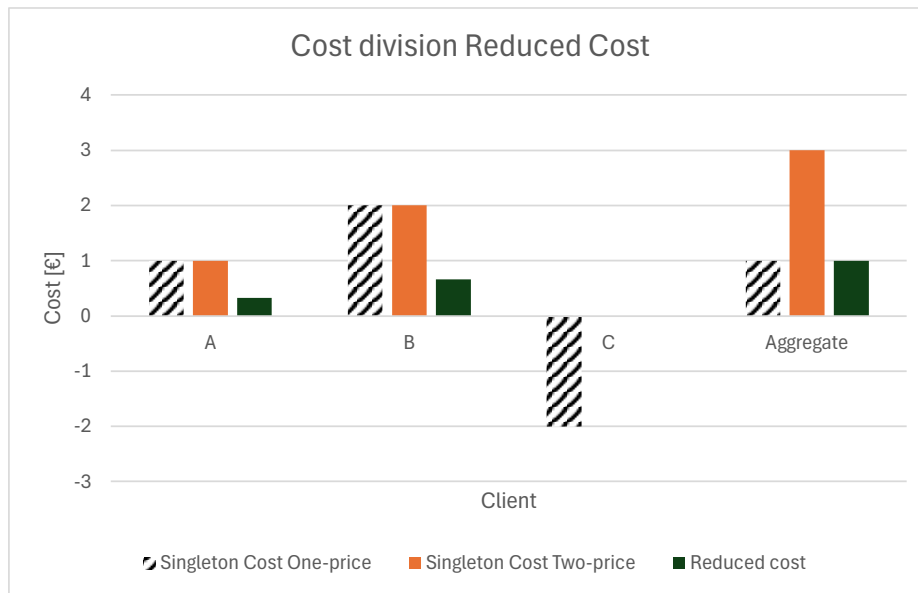


Error in thesis, k^{uni} is between 0 and imbalance spread

For cost:
$$x_i = \sum_{t \in T} \begin{cases} -k_t^{\text{uni}} \cdot |\Delta_{t,i}^{\text{Real}}|, & \text{if } v(\{i\}) = 0. \\ v_t(\{i\}), & \text{otherwise.} \end{cases}$$

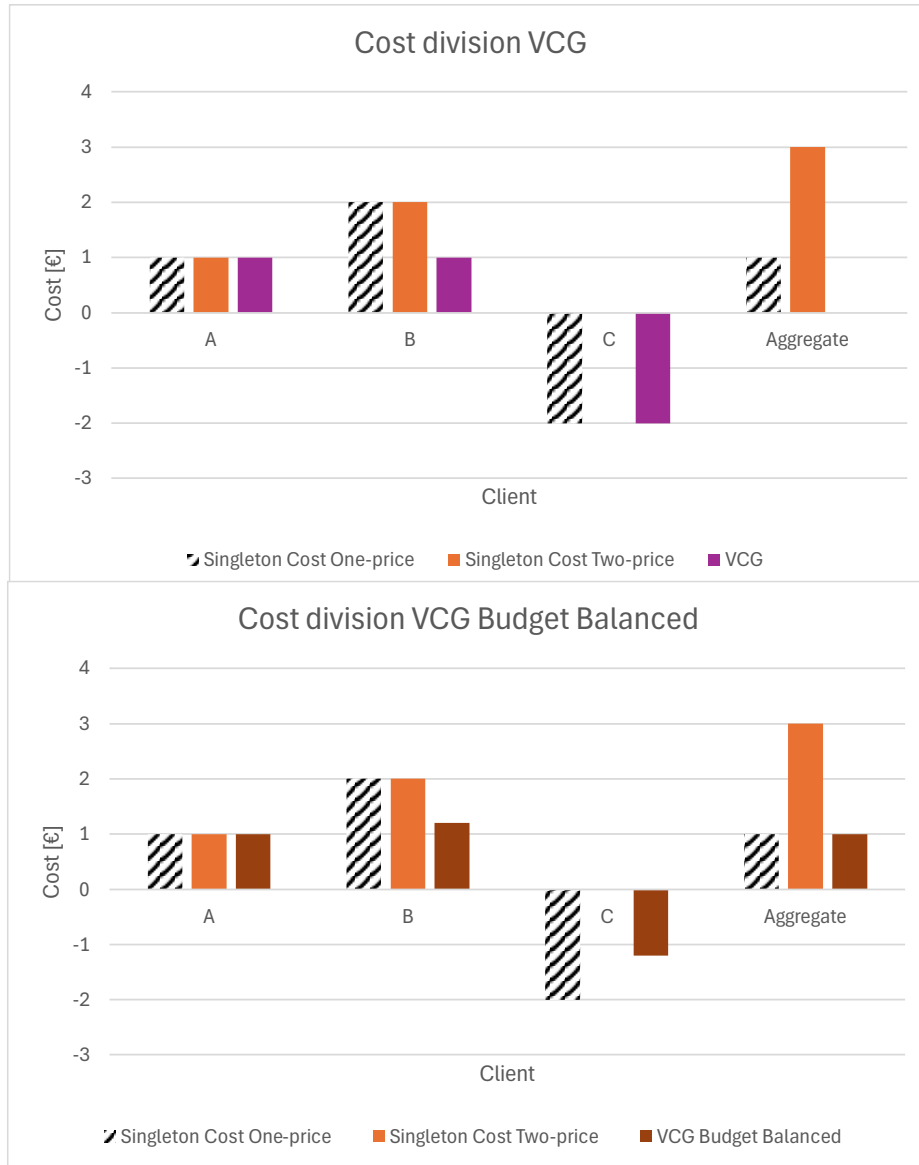
For risk:
$$x_i = \frac{1}{|\mathcal{T}_\alpha(J_{t,N})|} \sum_{t \in \mathcal{T}_\alpha(J_{t,N})} \begin{cases} -k_t^{\text{uni}} \cdot |\Delta_{t,i}^{\text{Real}}|, & \text{if } J_{t,i} = 0. \\ J_{t,i}, & \text{otherwise.} \end{cases}$$

Reduced Cost



$$x_i^N = \sum_{t \in T} \frac{v_t(\{i\})}{\sum_{j \in N} v_t(\{j\})} \cdot v_t(N)$$

VCG



$$\begin{aligned}
 x_i^N &= v_i(N) - \left(\sum_{j \in N \setminus \{i\}} v_j(N \setminus \{i\}) - \sum_{j \in N \setminus \{i\}} v_j(N) \right) \\
 &= v(N) - \sum_{j \in N \setminus \{i\}} v_j(N \setminus \{i\}) = M_i
 \end{aligned}$$

$$\min_{x_i} \sum_{i \in N} L(x_i, x_i^{\text{vick}})$$

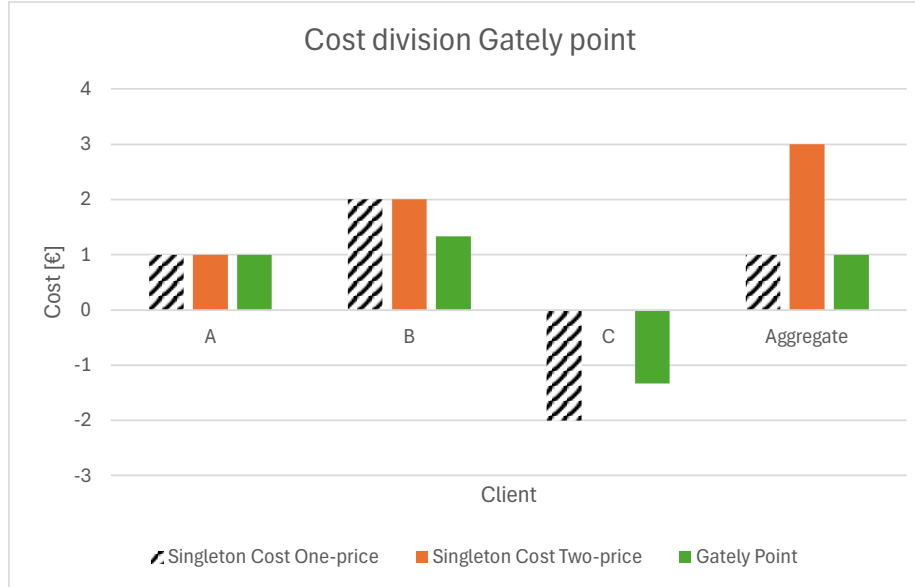
s.t.

$$\sum_{i \in N} x_i = v(N)$$

$$x_i \geq x_i^{\text{vick}}, \quad \forall i \in N$$

$$x_i \leq v(\{i\}), \quad \forall i \in N$$

Gately Point



- Full definition:

$$d_i(x) = \frac{\sum_{j \neq i} x_j - v(N \setminus \{i\})}{x_i - v(\{i\})} = \frac{v(N) - v(N \setminus \{i\}) - x_i}{x_i - v(\{i\})} = \frac{M_i - x_i}{x_i - v(\{i\})}$$

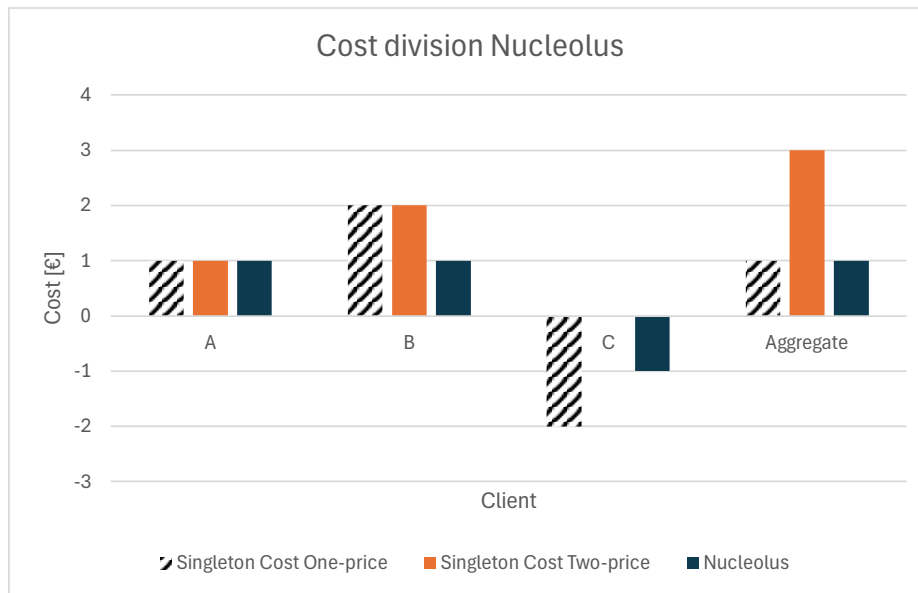
$$x^N = \arg \min_{x^N \in X^N} \left(\max_{i \in N} d_i(x_i^N) \right)$$

- Closed form:

$$d^* = \frac{\sum_{j \in N} M_j - v(N)}{v(N) - \sum_{j \in N} v_j}$$

$$x_i^N = \frac{d^* v(\{i\}) + M_i}{d + 1}$$

Nucleolus



$$\theta_i(x) = \epsilon(S_i, x), \text{ where } i = 1, \dots, 2^{|N|} - 1$$

$$x = \{y \in I \mid \theta(z) \succeq_l \theta(y) \forall z \in I\}$$

Stochastic bidding model

$$\min_{P_{t,S}^{DA}, \Delta_{w,t}, \Delta_{w,t}^+, \Delta_{w,t}^-} \sum_{u \in U} \sum_{w \in W} \pi^\lambda \pi^D \left(\sum_{t \in T} (\sigma_{u,t} \lambda_{u,t} \Delta_{w,t}^+ + (1 - \sigma_{u,t}) \lambda_{u,t} \Delta_{w,t}^-) \right) \quad (3.1)$$

s.t.

$$\Delta_{w,t} = \sum_{i \in S} P_{w,t,i}^D - \sum_{i \in S} P_{t,i}^G - P_{t,S}^{DA}, \quad \forall w \in W, t \in T \quad (3.2)$$

$$\Delta_{w,t}^+ - \Delta_{w,t}^- = \Delta_{w,t}, \quad \forall w \in W, t \in T \quad (3.3)$$

$$\Delta_{w,t}^+ \geq 0, \quad \forall w \in W, t \in T \quad (3.4)$$

$$\Delta_{w,t}^- \geq 0, \quad \forall w \in W, t \in T \quad (3.5)$$

Model error

	Uncertain PV Forecast	Perfect PV Forecast
Perfect Demand Forecast	18.1%	0%
Scenarios Demand Forecast	35.1%	26.0%
Noise Demand Forecast	22.9%	8.6%

Table 6.2: Weighted mean absolute percentage error for forecast type combinations